MIL-S-50389(MU)
21 October 1971

SUPERSEDING
FA-PD-MI-2631, Rev. 0
18 February 1965

MILITARY SPECIFICATION

SETTING MECHANISM ASSEMBLY (TRAINING)

1. SCOPE

1.1 This specification is for one type of manually operated, dial indicating, setting mechanism.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-10025	- Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F10534852	- Setting Mechanism Assembly (Trainer)

FSC 1390

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2.1 (Cont'd)

PUBLICATIONS

IEL-10534852 - Inspection Equipment List for Setting Mechanism Assembly (Trainer)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications except as provided for in 6.1.

3.2 Operating torque.-The setting mechanism shall operate and commence indicated readings when a torque of 6.0 inch pounds, maximum, is applied to the setting shaft in a clockwise direction.

3.3 Minimum life.-The setting mechanism shall operate without stopping the dials when the setting shaft has been subjected to 100 revolutions.

3.4 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.4.1 Plastic defects.-Parts made in accordance with drawings shall be extruded or injection molded. The finished parts shall be within the tolerances specified, shall have a smooth finish, and be free of voids, pin holes or other defects. Surface irregularities resulting from cutting off sprues and gates shall be removed by grinding and buffing until flush with adjacent surfaces.

3.4.2 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

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3.4.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.4.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.4.5 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Lot formation.-Each assembled lot of setting mechanisms shall be comprised of components each produced by one manufacturer, by one process, in accordance with the same drawing and revision and specifications and revisions, as applicable, and assembled on one assembly line.

4.2 First article sample.-When required, the first article sample shall be as specified in the contract and shall consist of the following:

5 - Setting mechanism assemblies and shall be subjected to the torque application and minimum life cycle as outlined in paragraphs 3.2 and 3.3 (see 4.4.5)

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

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4.2.1 Examination and test.-After inspection by the contractor and provisional acceptance at source by the Government, 5 setting mechanism assemblies shall be inspected, as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Acceptance criteria.-Failure of any unit to comply with the requirements of 3.2 or 3.3 as applicable, shall be cause for rejection of the first article sample.

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.2 Examination.-Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming materials shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Indicator Assembly No. 1, Drawing C10522902 covering a detail of Drawing F10534852.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Total length, max.	Gage
102.	Improper assembly	Visual
103.	Paint on gear teeth, or rider rim	Visual
104.	Metal defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
	None defined	

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4.3.2.2 Indicator Assemblies No. 2 and 3, Drawings C10522903 and C10522908 covering a detail of Drawing F10534852.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Sleeve not flush or below both surfaces of wheel	Visual
102. Paint on gear teeth or rider rim	Visual
MINOR	
None defined	

4.3.2.3 Gear and pinion assemblies No. 1, 2 and 3, Drawings C10522891, C10522885 and C10522888 covering a detail of Drawing F10534852.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of gear with datum	Gage
102. Perpendicularity of gear	Gage
103. Gear not seated against shoulder	Visual
104. Gear not secured properly	Visual
105. Burr	Visual
MINOR	
None defined	

4.3.2.4 Bracket Assembly Drawing D10522904 covering a detail of Drawing F10534852.

CRITICAL	
None defined	

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4.3.2.4 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	End play of counterwheel	Gage
102.	Intermittent rough spots when shaft is rotated	Manual
103.	Indicator No. 1 not shouldered properly	Visual
104.	Improper assembly	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
	None defined	

4.3.2.5 Housing (Trainer), Drawing F10534655 covering a detail of Drawing F10534852.

CRITICAL		
	None defined	
MAJOR		
101.	Diameter of four large holes in foot of housing, min.	Gage
102.	Location of four large holes in foot of housing	Gage
103.	Metal defective	Visual
MINOR		
201.	Evidence of poor workmanship	Visual

4.3.2.6 Setting Mechanism Assembly (Trainer), Drawing F10534852.

CRITICAL		
	None defined	

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Intermittent rough spots when shaft is rotated	Manual
102. Indicator numbers not clear or centered	Visual
103. Any part missing	Visual
104. Spring positioned on shaft, improperly	Visual
105. Marking missing or incorrect	Visual
106. Metal defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
None defined	

4.3.2.7 Packing and marking (Level C).-The number of specimens and acceptance criteria for packing and marking shall be in accordance with MIL-P-10025.

4.3.3 Inspection testing.-The following tests shall be performed in accordance with provisions of 4.4.

4.3.3.1 Protective finish (Nondestructive).-The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standards or Specifications listed on the drawing (See 4.4.1).

4.3.3.2 Static load and torque (Destructive).

4.3.3.2.1 Static load.-Ten (10) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (See 4.4.2).

Static Load	Drawings
No. 1 Gear and Pinion Assembly	C10522891
No. 2 Gear and Pinion Assembly	C10522885
No. 3 Gear and Pinion Assembly	C10522888

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4.3.3.2.2 Torque.-Ten (10) samples from each lot of the following subassemblies shall be subjected to the torque test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot subassemblies, as applicable (see 4.4.2).

Torque	Drawings
No. 1 Gear and Pinion Assembly	C10522891
No. 2 Gear and Pinion Assembly	C10522885
No. 3 Gear and Pinion Assembly	C10522888

4.3.3.3 Hardness (Nondestructive).-Ten (10) samples from each lot of the following components shall be tested for hardness. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot of components, as applicable (see 4.4.3).

Components	Drawings
Spring, Finger (Ratchet)	C10522900
Pin	B10522956
Shaft, Setting	C10524108
Gear, No. 2	C10522884
Gear, No. 3	C10522886
Gear, No. 1	C10522890
Pin	B10522955

4.3.3.4 Operating torque (ambient) (Nondestructive).-Each setting mechanism assembly produced shall be subjected to a torque application while at ambient temperature. Failure of any assembly to comply with the requirements of 3.2 shall be cause for rejection of the assembly (see 4.4.4).

4.3.4 Inspection equipment.-Index of Inspection Equipment Lists IEL No. 10523236 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with the applicable specification referenced on the drawing.

4.4.2 Static load and torque.-The test shall be performed using the values as specified on the applicable drawing. All samples tested shall be scrapped.

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4.4.3 Hardness.-Samples shall be tested in an area which is solidly supported so as to obtain valid readings. The median value of 3 hardness readings taken on each sample unit shall be used as the individual value. All components tested may be used for production.

4.4.4 Operating torque.-The torque shall be applied to the setting shaft in a clockwise direction.

4.4.5 Minimum life.-The setting shaft shall be rotated in a clockwise direction for 100 revolutions without stopping the dials. The min. and max. torque values shall be recorded. After breakdown and examinations, all components shall be scrapped.

5. PREPARATION FOR DELIVERY

5.1 Level A.-Not applicable.

5.2 Level B.-Not applicable.

5.3 Level C.-Packing and Marking.-Unless otherwise specified, packing and marking shall be in accordance with MIL-P-10025.

6. NOTES

6.1 Dimensions.

6.1.1 Compliance with tests (proving ground, ballistic performance, and shop) required by the specifications and drawings is mandatory. The contractor will not assume, nor does the Government guarantee, that all possible combinations permitted by the tolerance limits or the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerances and fits within the limits of the specifications and drawings that best suit his process needs and still satisfy the testing requirements.

6.1.2 Dimensions which are specifically referred to in the classification of defects, as listed in Section 4, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate fuze performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on fuze functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, and does not constitute authority to establish new dimensions or to relax efforts to achieve the required sizes within the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

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6.1.3 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6.3 Ordering data.-Procurement documents should specify the title, number and date of specification.

6.4 Supersession data.-This specification includes the requirements of Purchase Description FA-PD-MI-2631, Rev. 0 dated 18 February 1965.

Custodian:

Army - MU

Preparing activity

Army - MU(FA)

Project No. 1390-A803

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